

PONMUDI M.R.

RF / SDR & EMBEDDED SYSTEMS ENGINEER

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🌐 Portfolio

PROFESSIONAL SUMMARY

Electronics and Communication Engineering student with hands-on experience in embedded systems, RF/SDR signal processing, IoT, and custom PCB design. Proven track record building award-winning hardware systems spanning LoRa disaster communication, GPS-denied LEO satellite positioning, and custom PCB fabrication — taking projects from concept to national hackathon championships and defense/government research submissions (DRDO/iDEX, IEEE, Ministry of Education).

EDUCATION

B.E. Electronics and Communication Engineering

2022 - 2026 (2nd Year)

Bannari Amman Institute of Technology (BIT), Sathyamangalam, Tamil Nadu

TECHNICAL SKILLS

- **Programming:** C, Python, Arduino C/C++, Embedded C (learning), Bash
- **Embedded Systems:** ESP32, Arduino Mega, Raspberry Pi, Sensor Interfacing (SPI, I²C, UART)
- **RF & Wireless Comm:** LoRa (SX1278 / 433 MHz), SDR (RTL-SDR, SDR++), Wi-Fi, Doppler Shift / RF Signal Analysis
- **PCB & Hardware:** KiCad, Schematic Design, PCB Layout & Routing, Fabrication, Soldering
- **AI / Machine Learning:** YOLOv8 Signal Classification for RF Datasets, Synthetic Dataset Generation
- **Tools & OS:** Fedora Linux, Ubuntu, Git/GitHub, VS Code, Arduino IDE, Google Apps Script

KEY ENGINEERING PROJECTS

TribalNet – RF-Based Off-Grid Disaster Communication System

ESP32 • SX1278 LoRa (433 MHz) • MAX7219 Matrix • Embedded C

- Engineered a solar-powered, dual-node (RANGER ↔ VILLAGE) LoRa mesh communication network enabling emergency messaging in cellular-denied regions with MAX7219 LED matrix alert displays.
- 🏆 **1st Prize Winner (₹50,000)** at **NHIDE 2026 National Innovation Hackathon** (nhide2026.in | [LinkedIn Post](#)).
- Recognized as **IEEE YESIST12 2026 National Finalist** (Submission ID 3280); Patent Pending.
- Submitted to the **YUKTI National Innovation Repository** (Ministry of Education, Govt. of India).

Cognitive Opportunistic LEO-PNT Receiver

RTL-SDR • SDR++ • Python • YOLOv8 • Defense-Tech

- Developed a GPS-denied positioning system measuring Doppler shift telemetry from Low Earth Orbit (LEO) satellites (Orbcomm, NOAA, Iridium).
- Generated synthetic RF datasets & trained YOLOv8 classifier for signal types (BPSK, QPSK, SDPSK, TACAN).
- Submitted research proposal to **iDEX / DRDO** as part of Team 404's defense-tech research effort.

Auto Internet Login (nlogin)

Linux • Bash CLI • Git • Open-Source

- Built a Linux CLI automation tool for login/deauth on Cyberoam/Sophos captive portals. Published open-source.

Custom Police Siren Circuit & PCB Fabrication

KiCad • Hardware Prototyping • PCB Assembly

- Designed, routed, fabricated, and hand-assembled a custom police siren PCB, mastering hardware PCB workflows.

Kanaku Book – Digital Ledger PWA

Progressive Web App • Google Apps Script

- Built and published a payment ledger PWA for a family textile business with Google Apps Script backend.

HONORS & ACHIEVEMENTS

LEADERSHIP & AFFILIATIONS

🏆 **1st Prize (₹50,000)** – NHIDE 2026 Innovation Hackathon for TribalNet.

🏆 **IEEE YESIST12 2026** – National Finalist (Submission ID 3280).

🏆 **2nd Prize** – ChronoBIT'26 Hackathon (555 Timer + CD4026 Digital Clock).

📅 **Volunteer Coordinator** – Smart India Hackathon (Hardware Edition).

- **Team Lead, Team 404:** Leads a 3-member team (with Dinesh & Pratiksha) across hackathons.
- **Google Gemini Student Ambassador.**
- **Member:** IEEE Electron Devices Society, BIT-IECC, AViNYA Club.
- **Languages:** English, Tamil.